

Table 1: Sequential MAS results for Qwen3-8B. Acc. is accuracy (%), Token and Time values for Single, TextMAS, and Linear are taken directly from the reference figures, where Time is labeled Speed (A100). The remaining methods retain their H200 measurements. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
Sequential MAS Setting									
AIME24	Acc.	50.0	53.3	64.58	56.7	66.25	64.17	63.75	↑ 13.0
	Token	12891	38596	8607	8953	8548	8770	22342	↓ 77.9%
	Time	421	2808	229.8	688	225.6	225.8	549.0	× 4.5
AIME25	Acc.	46.7	53.3	48.75	53.3	49.58	50.42	49.17	↓ 2.9
	Token	14692	45088	9885	8699	10158	9830	24557	↓ 78.2%
	Time	450	3150	256.0	820	261.2	255.5	597.0	× 4.5
ARC-C	Acc.	91.0	94.6	94.54	94.4	94.80	95.56	94.45	↑ 1.0
	Token	846	2252	559	529	570	915	12970	↓ 74.7%
	Time	266	2059	3.7	703	3.7	4.5	65.3	× 3.9
ARC-E	Acc.	95.6	99.1	98.27	98.8	98.32	98.40	97.94	↓ 0.7
	Token	656	2085	468	490	472	822	10190	↓ 77.4%
	Time	404	3702	2.5	1759	2.5	3.2	48.8	× 4.4
GPQA	Acc.	39.9	43.4	49.87	45.5	48.74	48.61	49.24	↑ 5.3
	Token	6435	17986	3562	4571	3588	3846	13076	↓ 80.1%
	Time	813	5771	57.5	854	58.3	59.4	257.6	× 5.0
GSM8K	Acc.	81.1	92.3	91.58	93.8	91.74	92.19	—	↓ 0.1
	Token	1280	2324	628	860	642	1029	—	↓ 72.4%
	Time	449	1739	4.1	543	4.1	5.2	—	× 3.7
HumanEval+	Acc.	74.4	80.5	84.15	80.5	83.54	83.38	82.62	↑ 3.0
	Token	2507	4593	1668	1866	1637	1696	15055	↓ 64.4%
	Time	502	1619	18.9	497	18.8	21.5	112.4	× 2.8
MBPP+	Acc.	64.8	69.5	—	74.6	75.26	75.20	73.15	↑ 5.8
	Token	2053	3695	—	1164	1462	1528	14388	↓ 60.4%
	Time	1064	3628	—	1275	17.9	19.6	101.6	× 2.5
MedQA	Acc.	53.0	75.0	76.33	75.3	73.67	72.00	66.67	↓ 1.3
	Token	2098	4260	1872	1555	1139	1154	15972	↓ 73.3%
	Time	476	1923	17.0	928	12.5	13.6	111.1	× 3.7

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 2: Hierarchical MAS results for Qwen3-8B. Acc. is accuracy (%), Token and Time values for Single, TextMAS, and Linear are taken directly from the reference figures, where Time is labeled Speed (A100). The remaining methods retain their H200 measurements. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
Hierarchical MAS Setting									
AIME24	Acc.	50.0	53.3	—	53.3	60.00	65.83	—	↑ 12.5
	Token	12891	42629	—	7526	9926	9898	—	↓ 76.8%
	Time	421	3132	—	776	258.4	257.9	—	× 4.3
AIME25	Acc.	46.7	50.0	—	50.0	53.33	52.50	—	↑ 3.3
	Token	14692	53929	—	13230	11207	11300	—	↓ 79.2%
	Time	450	3488	—	616	289.1	291.9	—	× 4.8
ARC-C	Acc.	91.0	93.3	94.20	93.9	95.05	95.05	—	↑ 1.8
	Token	846	2854	606	344	603	822	—	↓ 78.9%
	Time	266	2034	3.9	714	3.8	4.3	—	× 4.7
ARC-E	Acc.	95.6	98.2	98.48	98.3	98.23	98.23	—	↑ 0.0
	Token	656	2237	494	308	495	721	—	↓ 77.9%
	Time	404	3619	2.6	1779	2.5	3.0	—	× 4.6
GPQA	Acc.	39.9	43.0	—	46.9	51.39	49.75	—	↑ 8.4
	Token	6435	22450	—	3395	3752	3964	—	↓ 83.3%
	Time	813	6108	—	798	59.9	61.5	—	× 6.0
GSM8K	Acc.	81.1	90.4	91.58	89.5	91.28	92.34	—	↑ 1.9
	Token	1280	2370	688	353	704	932	—	↓ 70.3%
	Time	449	1365	4.1	702	4.4	4.8	—	× 3.3
HumanEval+	Acc.	74.4	76.8	82.62	78.0	83.84	83.23	—	↑ 7.0
	Token	2507	8768	1688	1274	1681	1637	—	↓ 81.3%
	Time	502	1809	20.2	439	20.1	21.0	—	× 5.2
MBPP+	Acc.	64.8	71.9	70.97	72.2	72.75	72.16	73.21	↑ 0.8
	Token	2053	7703	1436	1264	1502	1460	12768	↓ 81.0%
	Time	1064	3898	20.3	1387	20.1	20.2	92.2	× 5.1
MedQA	Acc.	53.0	76.3	75.33	77.0	73.33	72.67	—	↓ 3.0
	Token	2098	6893	1391	1007	1351	1336	—	↓ 80.6%
	Time	476	3387	15.5	964	15.7	14.8	—	× 5.4

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 3: Sequential MAS results for Qwen3-14B. Acc. is accuracy (%), Token and Time values for Single, TextMAS, and Linear are taken directly from the reference figures, where Time is labeled Speed (A100). The remaining methods retain their H200 measurements. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
Sequential MAS Setting									
AIME24	Acc.	63.3	63.3	—	66.7	72.08	69.58	—	↑ 8.8
	Token	11263	32092	—	10593	8259	8489	—	↓ 74.3%
	Time	1018	4554	—	1149	255.6	283.4	—	× 3.9
AIME25	Acc.	56.7	60.0	—	63.3	56.67	60.00	49.58	↑ 0.0
	Token	11298	44618	—	11402	10266	10402	21403	↓ 77.0%
	Time	1040	5184	—	1473	335.5	350.6	575.5	× 4.3
ARC-C	Acc.	92.6	95.9	95.99	95.6	95.48	95.73	—	↓ 0.2
	Token	773	2985	463	426	461	544	—	↓ 84.6%
	Time	338	5125	4.3	1136	4.3	6.0	—	× 6.5
ARC-E	Acc.	97.2	99.0	98.48	99.4	98.82	98.57	—	↓ 0.2
	Token	608	1670	379	224	380	462	—	↓ 77.3%
	Time	551	9171	2.8	2124	2.7	4.5	—	× 4.4
GPQA	Acc.	48.5	51.5	—	52.0	55.81	54.42	53.54	↑ 4.3
	Token	5547	12676	—	5454	3009	2995	18528	↓ 76.4%
	Time	1043	9714	—	1475	59.4	67.2	222.9	× 4.2
GSM8K	Acc.	83.7	93.8	92.12	95.2	93.10	93.33	—	↓ 0.5
	Token	1118	3324	577	644	571	643	—	↓ 82.8%
	Time	536	3729	5.0	1952	4.7	6.7	—	× 5.8
HumanEval+	Acc.	76.8	81.1	86.89	86.5	88.11	86.43	90.70	↑ 7.0
	Token	2366	5934	1590	2042	1594	1607	19587	↓ 73.1%
	Time	1084	4062	24.1	1285	24.1	28.1	157.3	× 3.7
MBPP+	Acc.	68.5	72.8	78.17	75.7	78.11	79.43	—	↑ 6.6
	Token	1858	4971	1345	1621	1336	1346	—	↓ 73.1%
	Time	2410	8728	21.9	2400	21.5	25.5	—	× 3.7
MedQA	Acc.	64.7	80.3	81.00	80.7	81.67	79.33	79.00	↑ 1.4
	Token	1746	3444	1616	1841	922	912	22875	↓ 73.5%
	Time	1360	4142	19.0	1420	13.2	14.3	151.0	× 3.8

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 4: Hierarchical MAS results for Qwen3-14B. Acc. is accuracy (%), Token and Time values for Single, TextMAS, and Linear are taken directly from the reference figures, where Time is labeled Speed (A100). The remaining methods retain their H200 measurements. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
<i>Hierarchical MAS Setting</i>									
AIME24	Acc.	63.3	70.0	—	73.3	69.17	69.58	71.25	↓ 0.4
	Token	11263	29025	—	10230	8947	9017	23924	↓ 69.2%
	Time	1018	5718	—	1089	284.1	291.0	639.9	× 3.2
AIME25	Acc.	56.7	66.7	—	66.7	59.17	57.92	—	↓ 7.5
	Token	11298	50003	—	9527	10755	11122	—	↓ 78.5%
	Time	1040	6019	—	1056	346.9	365.7	—	× 4.6
ARC-C	Acc.	92.6	95.3	95.65	95.5	95.99	95.65	—	↑ 0.7
	Token	773	2167	434	295	424	477	—	↓ 80.4%
	Time	338	4283	4.0	1090	3.7	5.4	—	× 5.1
ARC-E	Acc.	97.2	98.3	98.70	98.7	98.78	98.23	—	↑ 0.5
	Token	608	2752	344	619	337	402	—	↓ 87.8%
	Time	551	7102	2.5	1884	2.4	4.0	—	× 8.2
GPQA	Acc.	48.5	52.0	—	53.0	54.80	55.81	54.55	↑ 3.8
	Token	5547	20931	—	3606	2769	2755	20909	↓ 86.8%
	Time	1043	9119	—	1458	61.5	63.6	287.8	× 7.6
GSM8K	Acc.	83.7	90.8	91.96	91.6	91.81	92.49	—	↑ 1.7
	Token	1118	3021	574	495	569	602	—	↓ 81.2%
	Time	536	3675	4.6	1631	4.6	6.0	—	× 5.3
HumanEval+	Acc.	76.8	84.1	88.41	86.6	87.04	86.74	91.16	↑ 2.9
	Token	2366	8114	1473	1512	1454	1430	21403	↓ 82.4%
	Time	1084	3988	23.8	1188	24.0	25.4	185.6	× 5.7
MBPP+	Acc.	68.5	73.0	76.92	73.8	78.04	78.31	—	↑ 5.3
	Token	1858	7458	1209	1187	1132	1182	—	↓ 84.8%
	Time	2410	9162	23.3	2507	21.6	23.7	—	× 6.6
MedQA	Acc.	64.7	78.0	81.33	78.3	79.00	77.67	48.33	↑ 1.0
	Token	1746	5473	870	899	860	875	23700	↓ 84.3%
	Time	1360	7591	12.1	1250	12.4	13.8	173.7	× 6.4

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.